

# Bach Phan-Tat

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Computational researcher working on interpretable, theory-driven modelling of change, with a primary focus on semantic and conceptual change. My work combines computational linguistics, statistical modelling, research software, and methodological evaluation, with developing interests in complex systems and time series.

## RESEARCH & PROFESSIONAL EXPERIENCE

**The Digital Humanities Institute, University of Sheffield**      **Sep 2026 – Nov 2026**  
*Visiting Research Scholar*      *Sheffield, England*  
*Project: Linguistic DNA*

- Refining the Linguistic DNA Concept Modelling method, developing statistical techniques for concept ranking and modelling the interaction between mutual information and frequency.
- Applying the method to the EEBO and ECCO corpora for large-scale analysis of diachronic conceptual change.
- Developing visualisation tools to support the exploration and interpretation of conceptual change.

**KU Leuven, QLVL Research Group**      **Sep 2024 – Present**  
*Doctoral Researcher, Computational Linguistics*      *Leuven, Belgium*  
*Project: Underlying Dimensions in Conceptual Change*

- Developed SynFlow, an open-source Python toolkit for multidimensional lexical semantic change analysis using structured linguistic patterns, distribution comparison, and incremental clustering of static embeddings.
- Built automated pipelines for LLM-assisted linguistic annotation and for cleaning, lemmatising, and parsing large corpora.
- Optimised computational workflows through parallel and batch processing, reducing processing time by 80%.
- Maintained NephoSem, a lexical-semantics toolkit developed by the QLVL research unit, and added support for other types of static embedding.

**Vbee (speech technology and conversational AI company)**      **Feb 2023 – Aug 2024**  
*Junior Research Engineer*      *Hanoi, Vietnam*  
*Project: HUST Chatbot (a conversational AI for Hanoi University of Science and Technology)*

- Developed and evaluated machine-learning and BERT-based models for chat domain classification, toxicity detection, and name-based classification, achieving over 85% accuracy on held-out evaluation sets.
- Designed data-sampling and human-evaluation procedures, applying statistical methods to assess dataset quality, annotation reliability, and model performance.
- Built an LLM-assisted chatbot evaluation pipeline, including evaluation criteria, prompt design, automated response scoring, and systematic analysis of model outputs.
- Developed language-data pipelines including web crawling, information extraction, LLM-assisted data generation, preprocessing, annotation, and quality control for text and speech datasets.
- Contributed to the development of linguistic resources, and rule-based components for speech and language tasks, including speech-to-text, text-to-speech, and part-of-speech parsing.

**what3words (location-mapping technology company)**      **Jul 2023 – Aug 2023**

*Remote Research Consultant*

*London, England*

*Project: Vietnamese Automatic Speech Recognition system (in collaboration with VinFast)*

- Conducted systematic linguistic error analysis of Vietnamese ASR outputs, categorising recurring errors involving phonetic confusions, lexical ambiguity, and contextual misrecognition.
- Translated error-analysis findings into data- and model-level recommendations.

**University of Stirling**

**Sep 2021 – Sep 2022**

*Master Student*

*Stirling, Scotland*

*Project: The structure of thời-gian*

- Crawled, cleaned, and compiled a specialised corpus of 500 randomly sampled Vietnamese news articles.
- Identified seven senses of the Vietnamese lexeme *thời gian* ('time'), their semantic structure and semantic development using the Principled Polysemy framework.

## EDUCATION

**KU Leuven**

**Sep 2024 – Present**

*PhD, Computational Linguistics*

*Leuven, Belgium*

Thesis: **Underlying Dimensions in Conceptual Change**

Thesis supervisors: Prof. Dr. Dirk Speelman, Prof. Dr. Dirk Geeraerts, Dr. Kris Heylen

**WorldQuant University**

**Oct 2025 – Present**

*MSc, Financial Engineering*

*Online*

Average grade (current): 84%

**University of Stirling**

**Sep 2021 – Sep 2022**

*MSc, English Language and Linguistics*

*Stirling, Scotland*

Thesis: **The structure of thời gian: A semantic analysis of the Vietnamese lexeme thời gian based on Principled Polysemy**

Thesis supervisor: Dr. Andrew Smith

Thesis grade: 80% (Distinction; class rank: 1st)

Average grade: 78% (Distinction; class rank: 1st)

**Academy of Finance**

**Aug 2016 – Aug 2020**

*BA, Corporate Finance*

*Hanoi, Vietnam*

Thesis grade: 9.2/10

Average grade: 7.15/10

## PUBLICATIONS

**Journal articles (peer-reviewed)**

- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, Dirk Speelman. 2026. Evaluating the Evaluator: Problems with SemEval-2020 Task 1 for Lexical Semantic Change Detection. *Journal of Open Humanities Data*.
- Catherine Wong, **Bach Phan-Tat**, Susan Fitzmaurice. 2026. Methods, Data, and Conceptual Change: Reflections from Two Quantitative Diachronic Case Studies. *Journal of Open Humanities Data*.

**Conference and workshop proceedings (peer-reviewed)**

- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. ReFRAME or Remain: Unsupervised Lexical Semantic Change Detection with Frame Semantics. *Proceedings of the 15th Joint Conference on Lexical and Computational Semantics (\*SEM 2026)*, ACL 2026.
- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. Transparent Semantic Change Detection with Dependency-Based Profiles. *Proceedings of the 6th International Workshop on Computational Approaches to Historical Language Change (LChange'26)*, EACL 2026.
- Khai Le-Duc, Tuyen Tran, **Bach Phan-Tat**, et al. 2025. MultiMed-ST: Large-scale Many-to-many Multilingual Medical Speech Translation. *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025)*.
- Khai Le-Duc, Phuc Phan, Tan-Hanh Pham, **Bach Phan-Tat**, Minh-Huong Ngo, Thanh Nguyen-Tang, Truong-Son Hy. 2025. MultiMed: Multilingual Medical Speech Recognition via Attention Encoder Decoder. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, Industry Track (ACL 2025)*. Top 15% of accepted papers.
- Khai Le-Duc, Khai-Nguyen Nguyen, **Bach Phan-Tat**, Le Duy, Long Vo-Dang, Truong-Son Hy. 2025. Sentiment Reasoning for Healthcare. *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics, Industry Track (ACL 2025)*. Oral presentation.

#### Book chapters (peer-reviewed)

- Sofía Aguilar-Valdez\*, **Bach Phan-Tat**\*, Dirk Speelman, Dirk Geeraerts, and Stefania Degaetano-Ortlieb. 2026. Discursive parallels of the chemical revolution: topic modelling and distributional analysis. In *Understanding Science with Large Language Models? Potentials for the History, Philosophy, and Sociology of Science* (open access, transcript). (Book chapter / edited volume). \*Equal contribution.

#### Pre-prints and under-review articles

- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, Dirk Speelman. 2026. SynFlow: A Multidimensional Diachronic Semantic Analysis Toolkit. (Preprint)
- Duc Cao-Dinh, Khai Le-Duc, Anh Dao, **Bach Phan-Tat**, Chris Ngo, Duy M. H. Nguyen, Nguyen X. Khanh, Thanh Nguyen-Tang. 2025. Audio-3DVG: Unified Audio - Point Cloud Fusion for 3D Visual Grounding. (Preprint).

## TALKS & PRESENTATIONS

#### Conference presentations

- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. Underlying Dimensions in Semantic Change. Digital Humanities Congress 2026.
- **Bach Phan-Tat**, Dirk Geeraerts, and Dirk Speelman. 2026. Conceptual Change during the Chemical Revolution: Air, Acid, and Water in the Royal Society Corpus. ICAME 47.
- **Bach Phan-Tat**, Kris Heylen, Dirk Geeraerts, Stefano De Pascale, and Dirk Speelman. 2026. From Parsers to Prompts: Combining AI and Linguistics for Interpretable Language Evolution. AI in Language Evolution Workshop, Evolang 2026.
- Khai Le-Duc, Khai-Nguyen Nguyen, **Bach Phan-Tat**, et al. 2024. Sentiment Reasoning for Healthcare. Advancements In Medical Foundation Models: Explainability, Robustness, Security, and Beyond Workshop, NeurIPS 2024.

#### Invited talks & seminars

- **Bach Phan-Tat**. 2026. Underlying Dimensions in Semantic Change. NLP & text mining seminar organised by Leuven.AI.

## ACADEMIC SERVICE & MENTORING

### Supervising & mentoring

- Advised a Master's thesis in Linguistics at Lancaster University, which is awarded a Distinction (2024).
- Mentored applicants in postgraduate and fellowship applications, with subsequent admissions to the Linguistics PhD programme at Lancaster University (2024), the YSEALI Fellowship (2024), the Stipendium Hungaricum Scholarship (2026).

### Reviewing

- Reviewer for Cognitive Linguistic Studies (2026), Linguists' Day, Linguistic Society of Belgium (2025).

### Organising

- Co-organised the PhD Day for the Department of Linguistics at KU Leuven. 2026.

## AWARDS & HONOURS

- **2026: First Place, Databricks Challenge** (US\$1,000), Hack-Nation's 4th Global AI Hackathon; 3,000+ participants from 300+ universities across 65+ countries.
- **2024: Marie Skłodowska-Curie Actions Doctoral Fellowship** (fully funded), Horizon Europe; awarded to 10 out of 284 applicants worldwide.
- **2022: Research-Based Learning Prize for the best Master's dissertation in Literature and Languages** (GBP 225), University of Stirling; the first recipient in the history of the Linguistics department.
- **2021: Postgraduate Scholarship** (GBP 4,000), University of Stirling.

## COMPUTATIONAL & RESEARCH SKILLS

### Programming & computing

- Python (NumPy, Pandas, Matplotlib, scikit-learn, spaCy, PyTorch, Hugging Face), R, Bash, LaTeX, MongoDB, SQL, Git/GitHub
- Parallel, batch, and GPU-enabled computing workflows

### Computational methods

- Exploratory data analysis, data preprocessing and (interactive) data visualisation
- Machine learning and deep learning for NLP, structured data, time-series and computer vision
- Quantitative research design, experimental design, and A/B testing
- LLM-assisted annotation and evaluation workflows

### Linguistics & corpus methods

- Corpus tools: SynFlow, CQPweb, #LancsBox, Sketch Engine, AntConc
- Linguistic analysis: phonetics, morphology, syntax, semantics
- Linguistic annotation, transcription, and corpus preparation

## SELECTED TRAINING

### Data science

- Applied Data Science Lab (WorldQuant University)
- Applied AI Lab: Deep Learning for Computer Vision (WorldQuant University)
- NeuroAI (Neuromatch Academy)

### **Neuroscience**

- Computational Neuroscience (Neuromatch Academy)

### **Philosophy**

- Philosophy and the Sciences: Introduction to the Philosophy of Cognitive Sciences (The University of Edinburgh)
- Philosophy and the Sciences: Introduction to the Philosophy of Physical Sciences (The University of Edinburgh)

## **LANGUAGES**

- Vietnamese: Native
- English: IELTS 8.5 – C2
- Dutch: A2